

The Effects of Group-Specific Employee Flows

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Abstract

This paper develops a theoretical framework for analyzing the effects of group-specific employee flows on workforce composition. The framework takes the group-specific employee flow rate as given and studies its implications for the size and representation of employee groups over time. The analysis distinguishes between absolute employee flows, relative employee flows, changes in group representation, and the effects of aggregate employment growth or contraction. Several propositions are derived concerning the conditions under which the representation of a group increases, decreases, or remains unchanged. The framework is descriptive and theoretical: it does not identify causal mechanisms and does not by itself establish discriminatory intent. Its purpose is to provide theoretical predictions that can subsequently be examined in an empirical paper.

The paper ends with “The End”

Introduction

The preceding paper introduced a framework for measuring group-specific employee flows. In that framework, the employee population is partitioned into identifiable groups and employee hiring and firing flows are measured separately for each group.

The principal group-specific employee flow rate is

$$c_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{t_2 - t_1},$$

where $h_g(t_1, t_2)$ denotes the number of employees in group g hired during the interval and $f_g(t_1, t_2)$ denotes the number of employees in group g fired during the interval.

The purpose of the present paper is different. It does not introduce another primary measurement. Instead, it asks:

What are the theoretical effects of group-specific employee flows on the composition and evolution of an employee population?

This distinction is important. A measure records an observable characteristic. A theory specifies what follows from that characteristic under a set of assumptions.

The Basic Workforce Model

Let the employee population consist of G groups indexed by

$$g \in \{1, 2, \dots, G\}.$$

Let

$$N_g(t)$$

denote the number of employees in group g at time t .

Total employment is

$$N(t) = \sum_{g=1}^G N_g(t).$$

The representation of group g is defined as

$$p_g(t) = \frac{N_g(t)}{N(t)}.$$

Therefore,

$$\sum_{g=1}^G p_g(t) = 1.$$

For the interval (t_1, t_2) , define

$$\Delta t = t_2 - t_1 > 0.$$

The group-specific net employee flow is

$$\Delta N_g = h_g(t_1, t_2) - f_g(t_1, t_2).$$

Consequently,

$$N_g(t_2) = N_g(t_1) + \Delta N_g.$$

Dividing by Δt gives the group-specific flow rate

$$c_g(t_1, t_2) = \frac{\Delta N_g}{\Delta t}.$$

Hence,

$$\Delta N_g = c_g(t_1, t_2) \Delta t.$$

This identity is central to the theoretical analysis.

Effect on Group Size

The first effect of the group-specific flow rate concerns the absolute size of the group.

Suppose that

$$c_g(t_1, t_2) > 0.$$

Then

$$\Delta N_g > 0,$$

and therefore

$$N_g(t_2) > N_g(t_1).$$

The group becomes larger in absolute employee numbers.

Similarly, if

$$c_g(t_1, t_2) < 0,$$

then

$$N_g(t_2) < N_g(t_1).$$

Finally, if

$$c_g(t_1, t_2) = 0,$$

then

$$N_g(t_2) = N_g(t_1).$$

Thus the first theoretical effect is immediate:

$$\boxed{\text{sign}(c_g(t_1, t_2)) = \text{sign}(N_g(t_2) - N_g(t_1))} \quad (1)$$

for a fixed positive interval length.

Effect on Total Employment

Summing group-specific flows across all groups gives

$$\sum_{g=1}^G \Delta N_g = \sum_{g=1}^G [h_g(t_1, t_2) - f_g(t_1, t_2)].$$

Therefore,

$$N(t_2) - N(t_1) = \sum_{g=1}^G \Delta N_g.$$

Dividing by Δt gives

$$\frac{N(t_2) - N(t_1)}{\Delta t} = \sum_{g=1}^G c_g(t_1, t_2).$$

Hence total employment growth is the sum of group-specific net employee flow rates. Define the aggregate employee flow rate by

$$c(t_1, t_2) = \frac{N(t_2) - N(t_1)}{\Delta t}.$$

Then

$$\boxed{c(t_1, t_2) = \sum_{g=1}^G c_g(t_1, t_2)}.$$

This establishes an aggregation property: aggregate net employment flow is exactly decomposable into group-specific net flows.

Effect on Group Representation

Absolute group growth does not necessarily imply an increase in group representation.

Recall that

$$p_g(t) = \frac{N_g(t)}{N(t)}.$$

Therefore,

$$p_g(t_2) = \frac{N_g(t_1) + \Delta N_g}{N(t_1) + \sum_{j=1}^G \Delta N_j}.$$

Using the group-specific flow rates,

$$p_g(t_2) = \frac{N_g(t_1) + c_g \Delta t}{N(t_1) + \left(\sum_{j=1}^G c_j \right) \Delta t}.$$

Thus the change in representation depends not only on the flow of group g , but also on the flows of all other groups.

This yields a fundamental distinction:

A group can grow in absolute size while simultaneously losing representation within the total workforce.

Representation Increase

We seek the condition under which

$$p_g(t_2) > p_g(t_1).$$

Using

$$p_g(t_1) = \frac{N_g(t_1)}{N(t_1)},$$

we obtain

$$\frac{N_g(t_1) + c_g \Delta t}{N(t_1) + \left(\sum_j c_j\right) \Delta t} > \frac{N_g(t_1)}{N(t_1)}.$$

Multiplying through by positive denominators gives

$$N(t_1)c_g \Delta t > N_g(t_1) \left(\sum_j c_j\right) \Delta t.$$

Since $\Delta t > 0$,

$$N(t_1)c_g > N_g(t_1) \sum_j c_j.$$

Using

$$p_g(t_1) = \frac{N_g(t_1)}{N(t_1)},$$

we obtain

$$\boxed{c_g > p_g(t_1) \sum_{j=1}^G c_j} \tag{2}$$

as the condition for the representation of group g to increase.
This is one of the principal theoretical results of the paper.

Representation Decrease

By the same argument,

$$p_g(t_2) < p_g(t_1)$$

if and only if

$$\boxed{c_g < p_g(t_1) \sum_{j=1}^G c_j}. \tag{3}$$

Consequently, the representation of a group is determined by its flow relative to the workforce-wide flow weighted by its initial representation.

Representation Stability

The representation of group g remains unchanged when

$$p_g(t_2) = p_g(t_1).$$

From the preceding derivation, this occurs when

$$\boxed{c_g = p_g(t_1) \sum_{j=1}^G c_j}. \quad (4)$$

This condition has an intuitive interpretation.

A group preserves its workforce share when its net flow is exactly equal to the flow that would be implied by maintaining its initial share of the total workforce.

The Neutral-Share Flow

Define the neutral-share flow for group g as

$$c_g^* = p_g(t_1) \sum_{j=1}^G c_j.$$

Then:

$$c_g > c_g^* \implies p_g(t_2) > p_g(t_1), \quad (5)$$

$$c_g < c_g^* \implies p_g(t_2) < p_g(t_1), \quad (6)$$

$$c_g = c_g^* \implies p_g(t_2) = p_g(t_1). \quad (7)$$

The quantity c_g^* is therefore the flow consistent with maintaining the group's existing representation. Define the representation pressure

$$\pi_g = c_g - c_g^*.$$

Then

$$\pi_g > 0$$

corresponds to upward representation pressure,

$$\pi_g < 0$$

corresponds to downward representation pressure, and

$$\pi_g = 0$$

corresponds to representation stability.

This quantity is derived from the original flow measure and is used here as a theoretical object rather than as a replacement for the primary measure.

Relative Employee Flow

The previous paper defined the normalized relative flow

$$q_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{N_g(t_1)(t_2 - t_1)}.$$

Using the definition of c_g ,

$$q_g = \frac{c_g}{N_g(t_1)}.$$

Thus

$$c_g = q_g N_g(t_1).$$

This provides an alternative interpretation of the theoretical effects.

The absolute flow c_g determines the change in group size, while q_g determines the net flow relative to the group's initial size.

Two groups may therefore have:

$$c_i > c_j$$

while simultaneously having

$$q_i < q_j.$$

This occurs whenever group i is sufficiently larger than group j at the beginning of the interval.

Differential Group Growth

Consider two groups i and j .

Their absolute net flows are

$$c_i \quad \text{and} \quad c_j.$$

Their normalized flows are

$$q_i = \frac{c_i}{N_i(t_1)}$$

and

$$q_j = \frac{c_j}{N_j(t_1)}.$$

The difference

$$c_i - c_j$$

therefore measures differential absolute expansion, while

$$q_i - q_j$$

measures differential expansion relative to group size.

These are distinct theoretical dimensions.

Consequently, an analysis of workforce effects should not infer relative group growth solely from absolute employee counts.

Aggregate Expansion

Suppose

$$\sum_{j=1}^G c_j > 0.$$

Then total employment is increasing.

In this case, a group can have

$$c_g > 0$$

and still experience

$$p_g(t_2) < p_g(t_1).$$

The necessary condition is

$$0 < c_g < p_g(t_1) \sum_j c_j.$$

Thus a growing group can lose representation when the rest of the workforce grows sufficiently rapidly.

Aggregate Contraction

Now suppose

$$\sum_{j=1}^G c_j < 0.$$

Then total employment is decreasing.

A group can nevertheless have

$$c_g < 0$$

while experiencing

$$p_g(t_2) > p_g(t_1).$$

This occurs when the group contracts less rapidly than the aggregate workforce.

Therefore, workforce representation and absolute employment can move in opposite directions under both expansion and contraction.

Special Case: Constant Total Employment

Suppose

$$\sum_{j=1}^G c_j = 0.$$

Then

$$N(t_2) = N(t_1).$$

The representation condition simplifies to

$$c_g > 0 \implies p_g(t_2) > p_g(t_1),$$

and

$$c_g < 0 \implies p_g(t_2) < p_g(t_1).$$

Thus, when total employment is constant, absolute group flow and representation change have the same sign.

This is a particularly transparent special case.

Gross Hiring and Firing Effects

The net flow rate alone does not reveal the gross magnitude of hiring and firing.

Recall

$$H_g = \frac{h_g}{\Delta t}$$

and

$$F_g = \frac{f_g}{\Delta t}.$$

Then

$$c_g = H_g - F_g.$$

Consequently, identical net flows can arise from different underlying employee-flow structures. For example,

$$H_g = 100, \quad F_g = 50$$

and

$$H_g = 20, \quad F_g = -30$$

would not be economically meaningful because firing counts cannot be negative. However, the valid comparison

$$(H_g, F_g) = (100, 50)$$

and

$$(H_g, F_g) = (20, -30)$$

is likewise invalid.

Instead, two valid flow structures such as

$$(H_g, F_g) = (100, 50)$$

and

$$(H_g, F_g) = (20, 0)$$

produce different gross flows but the same net flow,

$$c_g = 50.$$

The theoretical implication is that identical net flow need not imply identical organizational dynamics. Thus the theoretical framework distinguishes:

1. net employee flow;
2. gross hiring flow;
3. gross firing flow; and
4. workforce composition.

A Proposition on Composition

Proposition 1. For a group g , representation increases over an interval if and only if its group-specific employee flow rate exceeds its neutral-share flow.

Formally,

$$p_g(t_2) > p_g(t_1)$$

if and only if

$$c_g > p_g(t_1) \sum_j c_j.$$

Proof. The result follows directly from the derivation in Section 5. □

A Proposition on Absolute and Relative Effects

Proposition 2. Positive absolute group flow does not imply positive representation change.

Proof. Suppose

$$c_g > 0$$

and

$$\sum_j c_j > 0.$$

If

$$c_g < p_g(t_1) \sum_j c_j,$$

then Proposition 1 implies

$$p_g(t_2) < p_g(t_1).$$

Therefore positive absolute growth is compatible with declining representation. $\square$

A Proposition on Workforce Contraction

Proposition 3. During aggregate workforce contraction, negative group flow does not necessarily imply declining representation.

Proof. Suppose

$$\sum_j c_j < 0$$

and

$$c_g < 0.$$

If

$$c_g > p_g(t_1) \sum_j c_j,$$

then Proposition 1 implies

$$p_g(t_2) > p_g(t_1).$$

Therefore a group may contract in absolute size while increasing its share of total employment. $\square$

Interpretation of Differential Flows

Consider two groups i and j .

Differences in

$$c_i - c_j$$

describe differences in absolute net employee flow.

Differences in

$$q_i - q_j$$

describe differences in normalized net flow.

Differences in

$$p_i(t_2) - p_i(t_1)$$

describe differences in workforce representation.
 These quantities answer different questions.
 A theoretical analysis should therefore avoid collapsing them into a single concept.
 In particular,

$$c_i > c_j$$

does not necessarily imply

$$q_i > q_j,$$

and neither condition necessarily implies

$$p_i(t_2) - p_i(t_1) > p_j(t_2) - p_j(t_1).$$

The effect on representation depends on both group-specific and aggregate flows.

Implications for Organizational Analysis

The framework generates several classes of theoretical observations.

- First, an organization can expand while the representation of some groups falls.
 - Second, an organization can contract while the representation of some groups rises.
 - Third, a group can experience positive net hiring while losing workforce share.
 - Fourth, a group can experience negative net flow while gaining workforce share.
 - Fifth, identical net flows can conceal materially different gross hiring and firing processes.
- These results demonstrate that employee composition cannot be inferred from aggregate churn alone.

What the Theory Does Not Establish

The theoretical framework establishes relationships between employee flows, group sizes, and workforce representation.

It does not establish the causes of those flows.

In particular, differences between groups do not by themselves establish that the organization engaged in discrimination, preferential hiring, preferential firing, or any other causal mechanism.

Such conclusions require additional assumptions and evidence.

Potential explanations for observed differences may include differences in applicant pools, occupational composition, employee experience, qualification distributions, labor-market conditions, compensation, geographical allocation, business-unit structure, voluntary turnover, organizational expansion, organizational contraction, or other factors.

The present paper therefore stops at the theoretical implications of the measures.

Theoretical Predictions for Empirical Analysis

The theory yields a set of predictions that can subsequently be tested.

Condition	Predicted effect	Object
$c_g > 0$	Group size increases	N_g
$c_g < 0$	Group size decreases	N_g
$c_g > p_g \sum_j c_j$	Representation increases	p_g
$c_g < p_g \sum_j c_j$	Representation decreases	p_g
$c_g = p_g \sum_j c_j$	Representation unchanged	p_g
$\sum_j c_j > 0$	Total employment increases	N
$\sum_j c_j < 0$	Total employment decreases	N
$\sum_j c_j = 0$	Total employment unchanged	N

Table 1: Principal theoretical implications.

These predictions provide the bridge to the empirical paper.

Conclusion

The group-specific employee flow rate provides a basis for a theory of workforce composition.

The central result is that the effect of a group's employee flow on its representation depends on the group's flow relative to its representation-weighted share of aggregate employee flow.

The condition

$$c_g > p_g(t_1) \sum_j c_j$$

implies increasing representation,
while

$$c_g < p_g(t_1) \sum_j c_j$$

implies decreasing representation.

Equality produces representation stability.

The framework therefore distinguishes absolute group growth from relative workforce representation and demonstrates that the two may move in different directions.

This paper intentionally remains theoretical. It does not estimate the proposed relationships, identify causal mechanisms, or determine whether observed group differences constitute bias.

Those questions belong to the empirical stage of the research program.

The End